Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
multiply a polynomial by a monomial.			
multiply a binomial by a binomial.			
multiply a trinomial by a binomial.			

ⓑ *What does this checklist tell you about your mastery of this section? What steps will you take to improve?*

10.4 Divide Monomials

Learning Objectives

By the end of this section, you will be able to:

- Simplify expressions using the Quotient Property of Exponents
- Simplify expressions with zero exponents
- Simplify expressions using the Quotient to a Power Property
- Simplify expressions by applying several properties
- Divide monomials

☑ **BE PREPARED** 10.9 Before you get started, take this readiness quiz.

Simplify: $\frac{8}{24}$.

If you missed the problem, review Example 4.19.

☑ **BE PREPARED** 10.10 Simplify: $\left(2m^3\right)^5$.

If you missed the problem, review Example 10.23.

☑ **BE PREPARED** 10.11 Simplify: $\frac{12x}{12y}$.

If you missed the problem, review Example 4.23.

Simplify Expressions Using the Quotient Property of Exponents

Earlier in this chapter, we developed the properties of exponents for multiplication. We summarize these properties here.

Summary of Exponent Properties for Multiplication

If $a,\ b$ are real numbers and $m,\ n$ are whole numbers, then

Product Property	$a^m \cdot a^n = a^{m+n}$
Power Property	$(a^m)^n = a^{m \cdot n}$
Product to a Power	$(ab)^m = a^m b^m$

Now we will look at the exponent properties for division. A quick memory refresher may help before we get started. In Fractions you learned that fractions may be simplified by dividing out common factors from the numerator and denominator using the Equivalent Fractions Property. This property will also help us work with algebraic fractions—which are also quotients.

Equivalent Fractions Property

If $a,\ b,\ c$ are whole numbers where $b \neq 0,\ c \neq 0$, then

$$\frac{a}{b} = \frac{a \cdot c}{b \cdot c} \text{ and } \frac{a \cdot c}{b \cdot c} = \frac{a}{b}$$

As before, we'll try to discover a property by looking at some examples.

Consider	$\frac{x^5}{x^2}$	and	$\frac{x^2}{x^3}$
What do they mean?	$\frac{x\cdot x\cdot x\cdot x\cdot x}{x\cdot x}$		$\frac{x\cdot x}{x\cdot x\cdot x}$
Use the Equivalent Fractions Property.	$\frac{\cancel{x}\cdot\cancel{x}\cdot x\cdot x\cdot x}{\cancel{x}\cdot\cancel{x}\cdot 1}$		$\frac{\cancel{x}\cdot\cancel{x}\cdot 1}{\cancel{x}\cdot\cancel{x}\cdot x}$
Simplify.	x^3		$\frac{1}{x}$

Notice that in each case the bases were the same and we subtracted the exponents.

- When the larger exponent was in the numerator, we were left with factors in the numerator and 1 in the denominator, which we simplified.
- When the larger exponent was in the denominator, we were left with factors in the denominator, and 1 in the numerator, which could not be simplified.

We write:

$\frac{x^5}{x^2}$	$\frac{x^2}{x^3}$
x^{5-2}	$\frac{1}{x^{3-2}}$
x^3	$\frac{1}{x}$

Quotient Property of Exponents

If a is a real number, $a \neq 0$, and $m,\ n$ are whole numbers, then

$$\frac{a^m}{a^n} = a^{m-n}, \quad m > n \quad \text{and} \quad \frac{a^m}{a^n} = \frac{1}{a^{n-m}}, \quad n > m$$

A couple of examples with numbers may help to verify this property.

$\frac{3^4}{3^2} \stackrel{?}{=} 3^{4-2}$	$\frac{5^2}{5^3} \stackrel{?}{=} \frac{1}{5^{3-2}}$
$\frac{81}{9} \stackrel{?}{=} 3^2$	$\frac{25}{125} \stackrel{?}{=} \frac{1}{5^1}$
$9 = 9$✓	$\frac{1}{5} = \frac{1}{5}$✓

When we work with numbers and the exponent is less than or equal to 3, we will apply the exponent. When the exponent is greater than 3, we leave the answer in exponential form.

EXAMPLE 10.45

Simplify:

ⓐ $\frac{x^{10}}{x^8}$ ⓑ $\frac{2^9}{2^2}$

Solution

To simplify an expression with a quotient, we need to first compare the exponents in the numerator and denominator.

ⓐ

Since 10 > 8, there are more factors of x in the numerator.	$\frac{x^{10}}{x^8}$
Use the quotient property with $m > n$, $\frac{a^m}{a^n} = a^{m-n}$.	x^{10-8}
Simplify.	x^2

ⓑ

Since 9 > 2, there are more factors of 2 in the numerator.	$\frac{2^9}{2^2}$
Use the quotient property with $m > n$, $\frac{a^m}{a^n} = a^{m-n}$.	2^{9-2}
Simplify.	2^7

Notice that when the larger exponent is in the numerator, we are left with factors in the numerator.

TRY IT 10.89 Simplify:

ⓐ $\frac{x^{12}}{x^9}$ ⓑ $\frac{7^{14}}{7^5}$

TRY IT 10.90 Simplify:

ⓐ $\frac{y^{23}}{y^{17}}$ ⓑ $\frac{8^{15}}{8^7}$

EXAMPLE 10.46

Simplify:

ⓐ $\frac{b^{10}}{b^{15}}$ ⓑ $\frac{3^3}{3^5}$

Solution

To simplify an expression with a quotient, we need to first compare the exponents in the numerator and denominator.

ⓐ

Since 15 > 10, there are more factors of b in the denominator.	$\frac{b^{10}}{b^{15}}$
Use the quotient property with $n > m$, $\frac{a^m}{a^n} = \frac{1}{a^{n-m}}$.	$\frac{1}{b^{15-10}}$
Simplify.	$\frac{1}{b^5}$

ⓑ

Since 5 > 3, there are more factors of 3 in the denominator.	$\frac{3^3}{3^5}$
Use the quotient property with $n > m$, $\frac{a^m}{a^n} = \frac{1}{a^{n-m}}$.	$\frac{1}{3^{5-3}}$
Simplify.	$\frac{1}{3^2}$
Apply the exponent.	$\frac{1}{9}$

Notice that when the larger exponent is in the denominator, we are left with factors in the denominator and 1 in the numerator.

TRY IT 10.91 Simplify:

ⓐ $\frac{x^8}{x^{15}}$ ⓑ $\frac{12^{11}}{12^{21}}$

TRY IT 10.92 Simplify:

ⓐ $\frac{m^{17}}{m^{26}}$ ⓑ $\frac{7^8}{7^{14}}$

EXAMPLE 10.47

Simplify:

ⓐ $\frac{a^5}{a^9}$ ⓑ $\frac{x^{11}}{x^7}$

Solution

ⓐ

Since 9 > 5, there are more a's in the denominator and so we will end up with factors in the denominator.	$\frac{a^5}{a^9}$
Use the Quotient Property for $n > m$, $\frac{a^m}{a^n} = \frac{1}{a^{n-m}}$.	$\frac{1}{a^{9-5}}$
Simplify.	$\frac{1}{a^4}$

ⓑ

Notice there are more factors of x in the numerator, since 11 > 7. So we will end up with factors in the numerator.	$\frac{x^{11}}{x^7}$
Use the Quotient Property for $m > n$, $\frac{a^m}{a^n} = a^{n-m}$.	x^{11-7}
Simplify.	x^4

TRY IT 10.93 Simplify:

ⓐ $\frac{b^{19}}{b^{11}}$ ⓑ $\frac{z^5}{z^{11}}$

TRY IT 10.94 Simplify:

ⓐ $\frac{p^9}{p^{17}}$ ⓑ $\frac{w^{13}}{w^9}$

Simplify Expressions with Zero Exponents

A special case of the Quotient Property is when the exponents of the numerator and denominator are equal, such as an expression like $\frac{a^m}{a^m}$. From earlier work with fractions, we know that

$$\frac{2}{2}=1 \qquad \frac{17}{17}=1 \qquad \frac{-43}{-43}=1$$

In words, a number divided by itself is 1. So $\frac{x}{x}=1$, for any x $(x\neq 0)$, since any number divided by itself is 1.

The Quotient Property of Exponents shows us how to simplify $\frac{a^m}{a^n}$ when $m>n$ and when $n<m$ by subtracting exponents. What if $m=n$?

Now we will simplify $\frac{a^m}{a^m}$ in two ways to lead us to the definition of the **zero exponent**.

Consider first $\frac{8}{8}$, which we know is 1.

	$\frac{8}{8}=1$
Write 8 as 2^3.	$\frac{2^3}{2^3}=1$
Subtract exponents.	$2^{3-3}=1$
Simplify.	$2^0=1$

In general, for $a\neq 0$:

$\frac{a^m}{a^m}$	$\frac{a^m}{a^m}$
a^{m-m}	$\frac{\overbrace{a\cdot a\cdot \ldots \cdot a}^{m \text{ factors}}}{\underbrace{a\cdot a\cdot \ldots \cdot a}_{m \text{ factors}}}$
a^0	1

We see $\frac{a^m}{a^n}$ simplifies to a a^0 and to 1. So $a^0=1$.

Zero Exponent

If a is a non-zero number, then $a^0=1$.

Any nonzero number raised to the zero power is 1.

In this text, we assume any variable that we raise to the zero power is not zero.

EXAMPLE 10.48

Simplify:

ⓐ 12^0 ⓑ y^0

Solution

The definition says any non-zero number raised to the zero power is 1.

ⓐ

	12^0
Use the definition of the zero exponent.	1

ⓑ

	y^0
Use the definition of the zero exponent.	1

TRY IT 10.95 Simplify:

ⓐ 17^0 ⓑ m^0

TRY IT 10.96 Simplify:

ⓐ k^0 ⓑ 29^0

Now that we have defined the zero exponent, we can expand all the Properties of Exponents to include whole number exponents.

What about raising an expression to the zero power? Let's look at $(2x)^0$. We can use the product to a power rule to rewrite this expression.

	$(2x)^0$
Use the Product to a Power Rule.	2^0x^0
Use the Zero Exponent Property.	$1 \cdot 1$
Simplify.	1

This tells us that any non-zero expression raised to the zero power is one.

EXAMPLE 10.49

Simplify: $(7z)^0$.

Solution

	$(7z)^0$
Use the definition of the zero exponent.	1

TRY IT 10.97 Simplify: $(-4y)^0$.

TRY IT 10.98 Simplify: $\left(\frac{2}{3}x\right)^0$.

EXAMPLE 10.50

Simplify:

ⓐ $\left(-3x^2y\right)^0$ ⓑ $-3x^2y^0$

Solution

ⓐ

The product is raised to the zero power.	$(-3x^2y)^0$
Use the definition of the zero exponent.	1

ⓑ

Notice that only the variable y is being raised to the zero power.	$-3x^2y^0$
Use the definition of the zero exponent.	$-3x^2 \cdot 1$
Simplify.	$-3x^2$

TRY IT 10.99 Simplify:

ⓐ $(7x^2y)^0$ ⓑ $7x^2y^0$

TRY IT 10.100 Simplify:

ⓐ $-23x^2y^0$ ⓑ $(-23x^2y)^0$

Simplify Expressions Using the Quotient to a Power Property

Now we will look at an example that will lead us to the Quotient to a Power Property.

	$\left(\frac{x}{y}\right)^3$
This means	$\frac{x}{y} \cdot \frac{x}{y} \cdot \frac{x}{y}$
Multiply the fractions.	$\frac{x \cdot x \cdot x}{y \cdot y \cdot y}$
Write with exponents.	$\frac{x^3}{y^3}$

Notice that the exponent applies to both the numerator and the denominator.

We see that $\left(\frac{x}{y}\right)^3$ is $\frac{x^3}{y^3}$.

We write: $\left(\frac{x}{y}\right)^3$

$\frac{x^3}{y^3}$

This leads to the Quotient to a Power Property for Exponents.

Quotient to a Power Property of Exponents

If a and b are real numbers, $b \neq 0$, and m is a counting number, then

$$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

To raise a fraction to a power, raise the numerator and denominator to that power.

An example with numbers may help you understand this property:

$$\left(\frac{2}{3}\right)^3 \stackrel{?}{=} \frac{2^3}{3^3}$$
$$\frac{2}{3}\cdot\frac{2}{3}\cdot\frac{2}{3} \stackrel{?}{=} \frac{8}{27}$$
$$\frac{8}{27} = \frac{8}{27}\checkmark$$

EXAMPLE 10.51

Simplify:

ⓐ $\left(\frac{5}{8}\right)^2$ ⓑ $\left(\frac{x}{3}\right)^4$ ⓒ $\left(\frac{y}{m}\right)^3$

Solution

ⓐ

	$\left(\frac{5}{8}\right)^2$
Use the Quotient to a Power Property, $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$.	$\frac{5^2}{8^2}$
Simplify.	$\frac{25}{64}$

ⓑ

	$\left(\frac{x}{3}\right)^4$
Use the Quotient to a Power Property, $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$.	$\frac{x^4}{3^4}$
Simplify.	$\frac{x^4}{81}$

ⓒ

	$\left(\frac{y}{m}\right)^3$
Raise the numerator and denominator to the third power.	$\frac{y^3}{m^3}$

TRY IT 10.101 Simplify:

ⓐ $\left(\frac{7}{9}\right)^2$ ⓑ $\left(\frac{y}{8}\right)^3$ ⓒ $\left(\frac{p}{q}\right)^6$

TRY IT 10.102 Simplify:

ⓐ $\left(\frac{1}{8}\right)^2$ ⓑ $\left(\frac{-5}{m}\right)^3$ ⓒ $\left(\frac{r}{s}\right)^4$

Simplify Expressions by Applying Several Properties

We'll now summarize all the properties of exponents so they are all together to refer to as we simplify expressions using several properties. Notice that they are now defined for whole number exponents.

Summary of Exponent Properties

If $a,\ b$ are real numbers and m, n are whole numbers, then

Product Property	$a^m \cdot a^n = a^{m+n}$
Power Property	$(a^m)^n = a^{m \cdot n}$
Product to a Power Property	$(ab)^m = a^m b^m$
Quotient Property	$\frac{a^m}{a^n} = a^{m-n}, a \neq 0,\ m > n$
	$\frac{a^m}{a^n} = \frac{1}{a^{n-m}}, a \neq 0,\ n > m$
Zero Exponent Definition	$a^0 = 1,\ a \neq 0$
Quotient to a Power Property	$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m},\ b \neq 0$

EXAMPLE 10.52

Simplify: $\frac{(x^2)^3}{x^5}$.

Solution

	$\frac{(x^2)^3}{x^5}$
Multiply the exponents in the numerator, using the Power Property.	$\frac{x^6}{x^5}$
Subtract the exponents.	x

TRY IT 10.103 Simplify: $\frac{(a^4)^5}{a^9}$.

TRY IT 10.104 Simplify: $\frac{\left(b^5\right)^6}{b^{11}}$.

EXAMPLE 10.53

Simplify: $\frac{m^8}{(m^2)^4}$.

Solution

	$\frac{m^8}{(m^2)^4}$
Multiply the exponents in the numerator, using the Power Property.	$\frac{m^8}{m^8}$
Subtract the exponents.	m^0
Zero power property	1

TRY IT 10.105 Simplify: $\frac{k^{11}}{(k^3)^3}$.

TRY IT 10.106 Simplify: $\frac{d^{23}}{(d^4)^6}$.

EXAMPLE 10.54

Simplify: $\left(\frac{x^7}{x^3}\right)^2$.

Solution

	$\left(\frac{x^7}{x^3}\right)^2$
Remember parentheses come before exponents, and the bases are the same so we can simplify inside the parentheses. Subtract the exponents.	$\left(x^{7-3}\right)^2$
Simplify.	$\left(x^4\right)^2$
Multiply the exponents.	x^8

TRY IT 10.107 Simplify: $\left(\frac{f^{14}}{f^8}\right)^2$.

TRY IT 10.108 Simplify: $\left(\frac{b^6}{b^{11}}\right)^2$.

EXAMPLE 10.55

Simplify: $\left(\frac{p^2}{q^5}\right)^3$.

Solution

Here we cannot simplify inside the parentheses first, since the bases are not the same.

	$\left(\frac{p^2}{q^5}\right)^3$
Raise the numerator and denominator to the third power using the Quotient to a Power Property, $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$	$\frac{\left(p^2\right)^3}{\left(q^5\right)^3}$
Use the Power Property, $(a^m)^n = a^{m \cdot n}$.	$\frac{p^6}{q^{15}}$

TRY IT 10.109 Simplify: $\left(\frac{m^3}{n^8}\right)^5$.

TRY IT 10.110 Simplify: $\left(\frac{t^{10}}{u^7}\right)^2$.

EXAMPLE 10.56

Simplify: $\left(\frac{2x^3}{3y}\right)^4$.

Solution

	$\left(\frac{2x^3}{3y}\right)^4$
Raise the numerator and denominator to the fourth power using the Quotient to a Power Property.	$\frac{\left(2x^3\right)^4}{(3y)^4}$
Raise each factor to the fourth power, using the Power to a Power Property.	$\frac{2^4\left(x^3\right)^4}{3^4 y^4}$
Use the Power Property and simplify.	$\frac{16x^{12}}{81y^4}$

TRY IT 10.111 Simplify: $\left(\frac{5b}{9c^3}\right)^2$.

TRY IT 10.112 Simplify: $\left(\frac{4p^4}{7q^5}\right)^3$.

EXAMPLE 10.57

Simplify: $\frac{(y^2)^3(y^2)^4}{(y^5)^4}$.

Solution

	$\frac{(y^2)^3(y^2)^4}{(y^5)^4}$
Use the Power Property.	$\frac{\left(y^6\right)\left(y^8\right)}{y^{20}}$
Add the exponents in the numerator, using the Product Property.	$\frac{y^{14}}{y^{20}}$
Use the Quotient Property.	$\frac{1}{y^6}$

TRY IT 10.113 Simplify: $\frac{(y^4)^4(y^3)^5}{(y^7)^6}$.

TRY IT 10.114 Simplify: $\frac{(3x^4)^2(x^3)^4}{(x^5)^3}$.

Divide Monomials

We have now seen all the properties of exponents. We'll use them to divide monomials. Later, you'll use them to divide polynomials.

EXAMPLE 10.58

Find the quotient: $56x^5 \div 7x^2$.

Solution

	$56x^5 \div 7x^2$
Rewrite as a fraction.	$\frac{56x^5}{7x^2}$
Use fraction multiplication to separate the number part from the variable part.	$\frac{56}{7} \cdot \frac{x^5}{x^2}$
Use the Quotient Property.	$8x^3$

TRY IT 10.115 Find the quotient: $63x^8 \div 9x^4$.

TRY IT 10.116 Find the quotient: $96y^{11} \div 6y^8$.

When we divide monomials with more than one variable, we write one fraction for each variable.

EXAMPLE 10.59

Find the quotient: $\frac{42x^2y^3}{-7xy^5}$.

Solution

	$\frac{42x^2y^3}{-7xy^5}$
Use fraction multiplication.	$\frac{42}{-7} \cdot \frac{x^2}{x} \cdot \frac{y^3}{y^5}$
Simplify and use the Quotient Property.	$-6 \cdot x \cdot \frac{1}{y^2}$
Multiply.	$-\frac{6x}{y^2}$

TRY IT 10.117 Find the quotient: $\frac{-84x^8y^3}{7x^{10}y^2}$.

TRY IT 10.118 Find the quotient: $\frac{-72a^4b^5}{-8a^9b^5}$.

EXAMPLE 10.60

Find the quotient: $\frac{24a^5b^3}{48ab^4}$.

Solution

	$\frac{24a^5b^3}{48ab^4}$
Use fraction multiplication.	$\frac{24}{48} \cdot \frac{a^5}{a} \cdot \frac{b^3}{b^4}$
Simplify and use the Quotient Property.	$\frac{1}{2} \cdot a^4 \cdot \frac{1}{b}$
Multiply.	$\frac{a^4}{2b}$

TRY IT 10.119 Find the quotient: $\frac{16a^7b^6}{24ab^8}$.

TRY IT 10.120 Find the quotient: $\frac{27p^4q^7}{-45p^{12}q}$.

Once you become familiar with the process and have practiced it step by step several times, you may be able to simplify a fraction in one step.

EXAMPLE 10.61

Find the quotient: $\frac{14x^7y^{12}}{21x^{11}y^6}$.

Solution

	$\frac{14x^7y^{12}}{21x^{11}y^6}$
Simplify and use the Quotient Property.	$\frac{2y^6}{3x^4}$

Be very careful to simplify $\frac{14}{21}$ by dividing out a common factor, and to simplify the variables by subtracting their exponents.

TRY IT 10.121 Find the quotient: $\frac{28x^5y^{14}}{49x^9y^{12}}$.

TRY IT 10.122 Find the quotient: $\frac{30m^5n^{11}}{48m^{10}n^{14}}$.

In all examples so far, there was no work to do in the numerator or denominator before simplifying the fraction. In the next example, we'll first find the product of two monomials in the numerator before we simplify the fraction.

EXAMPLE 10.62

Find the quotient: $\frac{(3x^3y^2)(10x^2y^3)}{6x^4y^5}$.

Solution

Remember, the fraction bar is a grouping symbol. We will simplify the numerator first.

	$\frac{(3x^3y^2)(10x^2y^3)}{6x^4y^5}$
Simplify the numerator.	$\frac{30x^5y^5}{6x^4y^5}$
Simplify, using the Quotient Rule.	$5x$

TRY IT 10.123 Find the quotient: $\frac{(3x^4y^5)(8x^2y^5)}{12x^5y^8}$.

TRY IT 10.124 Find the quotient: $\frac{(-6a^6b^9)(-8a^5b^8)}{-12a^{10}b^{12}}$.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Simplify a Quotient (https://openstaxcollege.org/l/24simpquot)
Zero Exponent (https://openstaxcollege.org/l/zeroexponent)
Quotient Rule (https://openstaxcollege.org/l/24quotientrule)
Polynomial Division (https://openstaxcollege.org/l/24polyndivision)

Polynomial Division 2 (https://openstaxcollege.org/l/24polydivision2)

SECTION 10.4 EXERCISES

Practice Makes Perfect

Simplify Expressions Using the Quotient Property of Exponents

In the following exercises, simplify.

219. $\frac{4^8}{4^2}$

220. $\frac{3^{12}}{3^4}$

221. $\frac{x^{12}}{x^3}$

222. $\frac{u^9}{u^3}$

223. $\frac{r^5}{r}$

224. $\frac{y^4}{y}$

225. $\frac{y^4}{y^{20}}$

226. $\frac{x^{10}}{x^{30}}$

227. $\frac{10^3}{10^{15}}$

228. $\frac{r^2}{r^8}$

229. $\frac{a}{a^9}$

230. $\frac{2}{2^5}$

Simplify Expressions with Zero Exponents

In the following exercises, simplify.

231. 5^0

232. 10^0

233. a^0

234. x^0

235. -7^0

236. -4^0

237. ⓐ $(10p)^0$ ⓑ $10p^0$

238. ⓐ $(3a)^0$ ⓑ $3a^0$

239. ⓐ $\left(-27x^5y\right)^0$ ⓑ $-27x^5y^0$

240. ⓐ $\left(-92y^8z\right)^0$ ⓑ $-92y^8z^0$

241. ⓐ 15^0 ⓑ 15^1

242. ⓐ -6^0 ⓑ -6^1

243. $2 \cdot x^0 + 5 \cdot y^0$

244. $8 \cdot m^0 - 4 \cdot n^0$

Simplify Expressions Using the Quotient to a Power Property

In the following exercises, simplify.

245. $\left(\frac{3}{2}\right)^5$

246. $\left(\frac{4}{5}\right)^3$

247. $\left(\frac{m}{6}\right)^3$

248. $\left(\frac{p}{2}\right)^5$

249. $\left(\frac{x}{y}\right)^{10}$

250. $\left(\frac{a}{b}\right)^8$

251. $\left(\frac{a}{3b}\right)^2$

252. $\left(\frac{2x}{y}\right)^4$

Simplify Expressions by Applying Several Properties

In the following exercises, simplify.

253. $\frac{\left(x^2\right)^4}{x^5}$

254. $\frac{\left(y^4\right)^3}{y^7}$

255. $\frac{\left(u^3\right)^4}{u^{10}}$

256. $\frac{\left(y^2\right)^5}{y^6}$

257. $\frac{y^8}{\left(y^5\right)^2}$

258. $\frac{p^{11}}{\left(p^5\right)^3}$

259. $\frac{r^5}{r^4 \cdot r}$

260. $\frac{a^3 \cdot a^4}{a^7}$

261. $\left(\frac{x^2}{x^8}\right)^3$

262. $\left(\frac{u}{u^{10}}\right)^2$

263. $\left(\frac{a^4 \cdot a^6}{a^3}\right)^2$

264. $\left(\frac{x^3 \cdot x^8}{x^4}\right)^3$

265. $\frac{\left(y^3\right)^5}{\left(y^4\right)^3}$

266. $\frac{\left(z^6\right)^2}{\left(z^2\right)^4}$

267. $\frac{\left(x^3\right)^6}{\left(x^4\right)^7}$

268. $\frac{\left(x^4\right)^8}{\left(x^5\right)^7}$

269. $\left(\frac{2r^3}{5s}\right)^4$

270. $\left(\frac{3m^2}{4n}\right)^3$

271. $\left(\frac{3y^2 \cdot y^5}{y^{15} \cdot y^8}\right)^0$

272. $\left(\frac{15z^4 \cdot z^9}{0.3z^2}\right)^0$

273. $\frac{\left(r^2\right)^5\left(r^4\right)^2}{\left(r^3\right)^7}$

274. $\frac{\left(p^4\right)^2\left(p^3\right)^5}{\left(p^2\right)^9}$

275. $\frac{\left(3x^4\right)^3\left(2x^3\right)^2}{\left(6x^5\right)^2}$

276. $\frac{\left(-2y^3\right)^4\left(3y^4\right)^2}{\left(-6y^3\right)^2}$

Divide Monomials

In the following exercises, divide the monomials.

277. $48b^8 \div 6b^2$

278. $42a^{14} \div 6a^2$

279. $36x^3 \div \left(-2x^9\right)$

280. $20u^8 \div \left(-4u^6\right)$

281. $\frac{18x^3}{9x^2}$

282. $\frac{36y^9}{4y^7}$

283. $\frac{-35x^7}{-42x^{13}}$

284. $\frac{18x^5}{-27x^9}$

285. $\frac{18r^5s}{3r^3s^9}$

286. $\frac{24p^7q}{6p^2q^5}$

287. $\frac{8mn^{10}}{64mn^4}$

288. $\frac{10a^4b}{50a^2b^6}$

289. $\frac{-12x^4y^9}{15x^6y^3}$

290. $\frac{48x^{11}y^9z^3}{36x^6y^8z^5}$

291. $\frac{64x^5y^9z^7}{48x^7y^{12}z^6}$

292. $\frac{(10u^2v)(4u^3v^6)}{5u^9v^2}$

293. $\frac{(6m^2n)(5m^4n^3)}{3m^{10}n^2}$

294. $\frac{(6a^4b^3)(4ab^5)}{(12a^8b)(a^3b)}$

295. $\frac{(4u^5v^4)(15u^8v)}{(12u^3v)(u^6v)}$

Mixed Practice

296. ⓐ $24a^5 + 2a^5$
ⓑ $24a^5 - 2a^5$
ⓒ $24a^5 \cdot 2a^5$
ⓓ $24a^5 \div 2a^5$

297. ⓐ $15n^{10} + 3n^{10}$
ⓑ $15n^{10} - 3n^{10}$
ⓒ $15n^{10} \cdot 3n^{10}$
ⓓ $15n^{10} \div 3n^{10}$

298. ⓐ $p^4 \cdot p^6$ ⓑ $\left(p^4\right)^6$

299. ⓐ $q^5 \cdot q^3$ ⓑ $\left(q^5\right)^3$

300. ⓐ $\frac{y^3}{y}$ ⓑ $\frac{y}{y^3}$

301. ⓐ $\frac{z^6}{z^5}$ ⓑ $\frac{z^5}{z^6}$

302. $\left(8x^5\right)(9x) \div 6x^3$

303. $(4y)\left(12y^7\right) \div 8y^2$

304. $\frac{27a^7}{3a^3} + \frac{54a^9}{9a^5}$

305. $\frac{32c^{11}}{4c^5} + \frac{42c^9}{6c^3}$

306. $\frac{32y^5}{8y^2} - \frac{60y^{10}}{5y^7}$

307. $\frac{48x^6}{6x^4} - \frac{35x^9}{7x^7}$

308. $\frac{63r^6s^3}{9r^4s^2} - \frac{72r^2s^2}{6s}$

309. $\frac{56y^4z^5}{7y^3z^3} - \frac{45y^2z^2}{5y}$

Everyday Math

310. **Memory** One megabyte is approximately 10^6 bytes. One gigabyte is approximately 10^9 bytes. How many megabytes are in one gigabyte?

311. **Memory** One megabyte is approximately 10^6 bytes. One terabyte is approximately 10^{12} bytes. How many megabytes are in one terabyte?